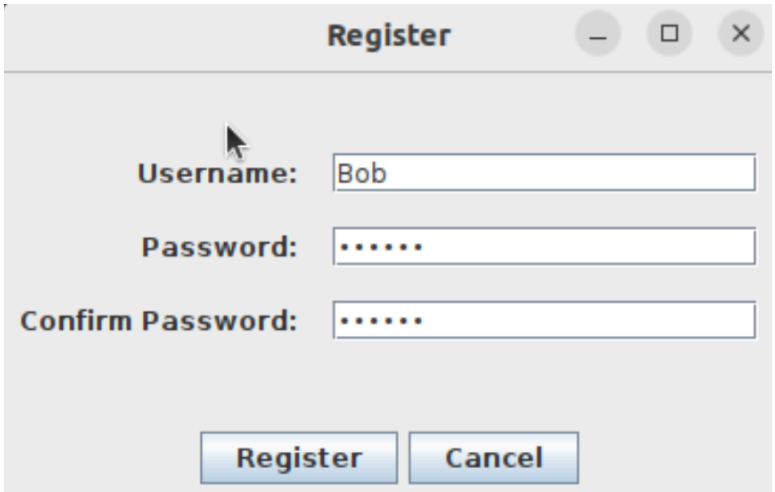
Database tables

```
mysql> describe online_users;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| username | varchar(50) | NO   | PRI | NULL    |       |
+----------+-------------+------+-----+---------+-------+
1 row in set (0.00 sec)
```

```
mysql> describe users;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| username         | varchar(50)  | NO   | PRI | NULL    |       |
| password         | varchar(255) | NO   |     | NULL    |       |
| num_games        | int          | NO   |     | 0       |       |
| num_wins         | int          | NO   |     | 0       |       |
| avg_time_to_win  | float        | NO   |     | 0       |       |
| leaderboard_rank | int          | NO   |     | 0       |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

Register



When register is pressed, the following steps are executed:

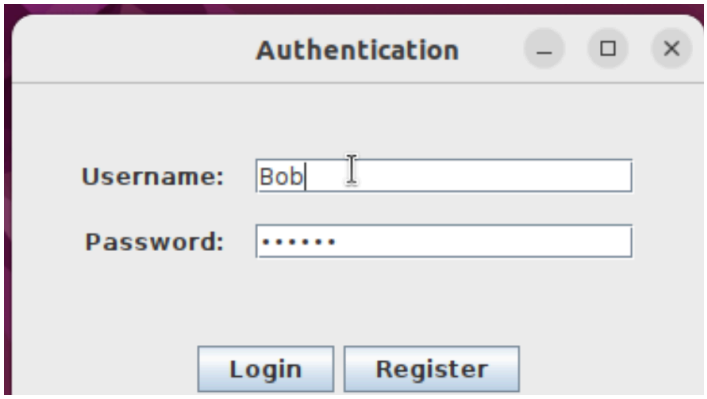
1. Check whether username already exists by querying users table. (READ) - This step is redundant, it is only here to provide clearer error messages for users.
2. If it doesn't, insert a new entry in users table. (INSERT)
   Registering with 'Bob' username
   Before

```
mysql> select * from users;
+----------+----------+-----------+----------+-----------------+------------------+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+----------+----------+-----------+----------+-----------------+------------------+
| a        | a        |         5 |        0 |               0 |                0 |
+----------+----------+-----------+----------+-----------------+------------------+
1 row in set (0.00 sec)
```

   After

```
mysql> select * from users;
+----------+----------+-----------+----------+-----------------+------------------+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+----------+----------+-----------+----------+-----------------+------------------+
| a        | a        |         5 |        0 |               0 |                0 |
| Bob      | 123456   |         0 |        0 |               0 |                0 |
+----------+----------+-----------+----------+-----------------+------------------+
```

Login



1. Check whether username and password are correct by querying users table (READ)
2. If correct, check whether user is already logged in by querying online_users table (READ) - This step is redundant, it is only here to provide clearer error messages for users.
3. If not logged in already, insert username into online_users table. (INSERT)

Logging in with 'Bob' username

Before

```
mysql> select * from online_users;
+----------+
| username |
+----------+
| a        |
+----------+
1 row in set (0.00 sec)
```

After

```
mysql> select * from online_users;
+----------+
| username |
+----------+
| a        |
| Bob      |
+----------+
2 rows in set (0.00 sec)
```

Logout

1. Remove username from online_users (DELETE)

Logging out as 'Bob' user

Before

```
mysql> select * from online_users;
+----------+
| username |
+----------+
| a        |
| Bob      |
+----------+
2 rows in set (0.00 sec)
```

After

```
mysql> select * from online_users;
+----------+
| username |
+----------+
| a        |
+----------+
1 row in set (0.00 sec)
```

Fetching user info (User Profile + Leaderboard)

User profile section

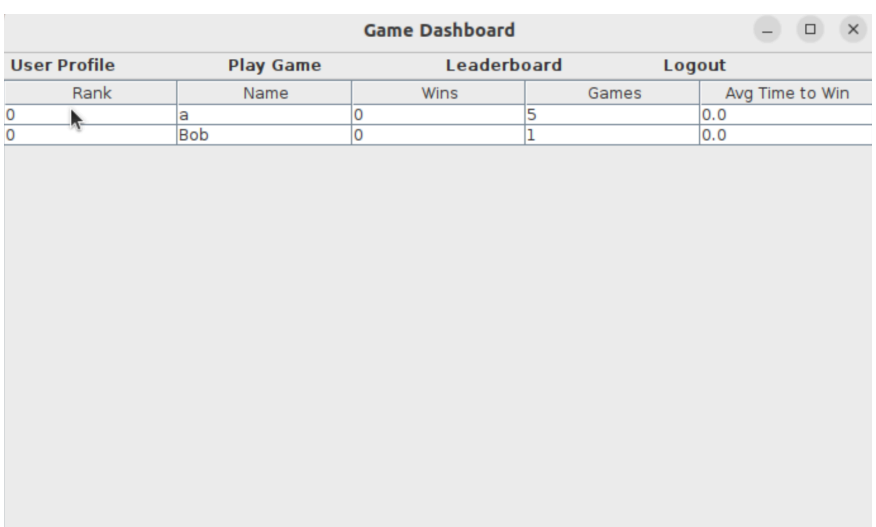
1. Query users table to fetch information for currently logged in user (READ)

```
mysql> select * from users where username = "Bob";
+----------+----------+-----------+----------+-----------------+------------------+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+----------+----------+-----------+----------+-----------------+------------------+
| Bob      | 123456   |         1 |        0 |               0 |                0 |
+----------+----------+-----------+----------+-----------------+------------------+
1 row in set (0.00 sec)
```

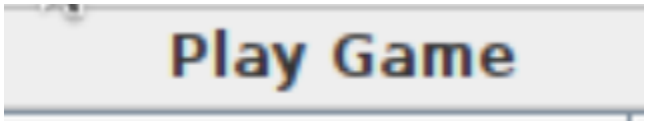
Leaderboard section

1. Query users table to fetch information for all users. (READ)

```
mysql> select * from users;
+----------+----------+-----------+----------+-----------------+------------------+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+----------+----------+-----------+----------+-----------------+------------------+
| a        | a        |         5 |        0 |               0 |                0 |
| Bob      | 123456   |         1 |        0 |               0 |                0 |
+----------+----------+-----------+----------+-----------------+------------------+
2 rows in set (0.00 sec)
```



Play game



For the sake of demonstrating the UPDATE operation, the ‘Play Game’ button will increment users’ num_games by 1 when clicked.

Pressing “Play Game” as Bob

Before

```
mysql> select num_games from users where username = "Bob";
+-----------+
| num_games |
+-----------+
|         1 |
+-----------+
1 row in set (0.00 sec)
```

After

```
mysql> select num_games from users where username = "Bob";
+-----------+
| num_games |
+-----------+
|         2 |
+-----------+
1 row in set (0.00 sec)
```

Consider whether transactions should be implemented in this application, and provide your reasoning.

As the application currently stands, there is no need for transactions to be implemented. The only scenario where it could be considered is during registration and login where we are checking specific conditions before inserting new entries (to either users or online_users). However, even if race conditions occur after the respective checks, the primary key constraint (username) will prevent entries with duplicate usernames from being inserted, thus the integrity of the database will still be maintained.

Consider the example for login: (steps will follow dist_assg2_part2)

1. User1 passes step 1
2. User2 passes step 1
3. User1 passes step 2
4. User2 passes step 2
5. User1 completes step 3 with username “Bob”.
6. User2 attempts step 3 with username “Bob”. The server will receive an error from the database, reminding that duplicate primary keys are prohibited.

A similar process happens for register when dealing with this type of race condition.